

FBISE CHEMISTRY CLASS 9

2nd Annual Examination 2026 • Best Guess Paper

Exam date	29 September 2026	Total marks	60
Predicted pattern	12 MCQs + 30 Short + 18 Long	Time	As per FBISE date sheet/paper instructions

Important limitation

This is an evidence-based prediction, not an official or leaked paper. No guess can guarantee the exact questions or full marks. Students should prepare every summative SLO; use this paper for final high-yield revision and timed practice.

Excluded from written-paper prediction

- Formative only: C-09-B-10 (probabilistic electron paths/uncertainty principle) and C-09-B-12 (atomic model as an aid).
- PBA only: C-09-D-12 to D-14 (water test with anhydrous CuSO_4 , purity by melting/boiling point, distilled vs tap water in practical chemistry).

How to Use This Guess

Priority	Meaning	Student action
A+	Very high probability; repeated across model/past papers or central to the current pattern.	Prepare complete answer, equation/diagram, reasons and units.
A	High probability; strongly supported by the assessment framework and recent variants.	Prepare at least a clear 3-mark response.
B	Reserve coverage for summative SLOs that can replace a predicted item.	Revise after all A+/A material.

Strongest evidence clusters

Rank	Cluster	Why it matters
A+	Atomic structure, isotopes and relative atomic mass	Appears repeatedly as proton/neutron determination, isotope definition/use and weighted-average calculation.
A+	Bonding and structure	Dot-and-cross diagrams, ionic/covalent classification, properties from structure, diamond/graphite and metallic bonding recur as long questions.
A+	Mole calculations and energetics	Mass-mole-particle conversion, bond-energy ΔH , activation energy and pathway diagrams recur in every source set.
A+	Periodicity and group properties	Group/period identification, periodic trends, Group I/VII and displacement reactions dominate MCQs, shorts and long questions.
A+	Acids/bases, equilibrium and redox	Definitions plus equations, equilibrium shifts, oxidation numbers and agents repeatedly receive 3-6 marks.
A	Environment, water and fertilizers	Pollutant effects, water treatment/scarcity and NPK/nitrogen fertilizers are regular short questions.
A+	Organic chemistry, alkanes and biomolecules	Functional groups/isomerism, substitution/preparation of alkanes and food sources/functions recur throughout.

SECTION A — Predicted MCQs

Marks: 12 • Attempt all questions. Choose the best option.

#	Question and options
1	The branch of chemistry that studies chemical processes in living organisms is: A. Astrochemistry B. Biochemistry C. Physical chemistry D. Analytical chemistry
2	Which statement correctly compares the states of matter? A. Gases have the highest density B. Solids are highly compressible C. Liquids are fluid but nearly incompressible D. Plasma is a solid allotrope
3	An isotope $^{37}_{17}\text{Cl}$ contains: A. 17 p, 17 n B. 17 p, 20 n C. 20 p, 17 n D. 37 p, 17 n
4	Which substance contains covalent bonding? A. NaF B. MgO C. H ₂ O D. CaCl ₂
5	The number of molecules in 2.0 mol CO ₂ is: A. 3.011×10^{23} B. 6.022×10^{23} C. 1.2044×10^{24} D. 2.0×10^{23}
6	The oxidation number of sulfur in SO ₄ ²⁻ is: A. +2 B. +4 C. +6 D. -2
7	For an endothermic reaction, the sign of ΔH is: A. Positive B. Negative C. Zero D. Always -1
8	For $\text{CoCl}_2 \cdot 6\text{H}_2\text{O} \rightleftharpoons \text{CoCl}_2 + 6\text{H}_2\text{O}$, adding water shifts equilibrium: A. To the right B. To the left C. In neither direction D. Randomly
9	An acid reacting with a metal carbonate produces: A. Salt + H ₂ B. Salt + CO ₂ + H ₂ O C. Salt + NH ₃ D. Salt only
10	Down Group I, which trend is generally correct? A. Atomic radius decreases B. Reactivity decreases C. Ionization energy increases D. Reactivity increases
11	A major source of SO ₂ in air is: A. Complete combustion of sulfur-containing fuels B. Evaporation of water C. Photosynthesis D. Chlorination
12	The functional group of carboxylic acids is: A. -OH B. -CHO C. -COOH D. -NH ₂

Answer key: 1-B • 2-C • 3-B • 4-C • 5-C • 6-C • 7-A • 8-B • 9-B • 10-D • 11-A • 12-C

MCQ Reserve Bank — Must Revise

These are the most likely replacement concepts if the exact predicted MCQ is changed.

Concept stem	Correct fact
Noble gases are unreactive because	their outermost shell is complete
The most compressible common state	gas
An atom forms a cation by	losing electron(s)
Relative atomic mass	weighted average relative to 1/12 of carbon-12
Oxidation number of a free element	zero
A catalyst	provides a lower-activation-energy pathway
Equilibrium in a closed system	has equal forward and reverse reaction rates
A strong acid/base	dissociates completely in water
Halogen reactivity down Group VII	decreases
Nitrogen in clean dry air	about 78%
CO and particulates form mainly by	incomplete combustion
Useful substance in natural water	dissolved oxygen
A saturated hydrocarbon	contains only C–C single bonds
Alkanes usually undergo	substitution reactions
General formula of alkanes	C_nH_{2n+2}
Proteins are abundant in	meat, eggs, pulses/legumes

SECTION B — Predicted Short Questions

Marks: 30 • Attempt one side of each numbered pair. Each part carries 3 marks.

A+	1(a) Identify the relevant branch of chemistry for: (a) Haber-process manufacture of NH_3 , (b) breakdown of glucose in the body, and (c) depletion of the ozone layer by CFCs.	OR	1(b) Define isotopes. For $^{23}_{11}\text{Na}$, $^{35}_{17}\text{Cl}$ and $^{14}_6\text{C}$, determine the numbers of protons and neutrons.
A+	2(a) Explain why solids have high density and very low compressibility whereas gases show the opposite behaviour.	OR	2(b) Differentiate among a solution, colloid and suspension; give one example of each.
A+	3(a) Draw electron dot-and-cross diagrams for the formation of NaF and H_2O , and identify the bond in each.	OR	3(b) Differentiate ionic and covalent compounds using melting point, solubility and electrical conductivity.
A+	4(a) Define mole. Calculate the number of moles in 100 g of aluminium and the number of particles present. (Ar Al = 27)	OR	4(b) Calculate the relative atomic mass of element X having isotopes X-35 (75%) and X-37 (25%).
A+	5(a) Define oxidation and reduction in terms of oxidation number. Identify the oxidizing and reducing agents in $\text{Fe}_2\text{O}_3 + 3\text{CO} \rightarrow 2\text{Fe} + 3\text{CO}_2$.	OR	5(b) Calculate the oxidation numbers of nitrogen in NH_4^+ and carbon in CO_3^{2-} ; show that their sums equal the ionic charges.
A+	6(a) Define activation energy. Explain how a catalyst changes activation energy and reaction rate.	OR	6(b) Using $\text{H}-\text{H} = 436$, $\text{Cl}-\text{Cl} = 242$ and $\text{H}-\text{Cl} = 431 \text{ kJ mol}^{-1}$, calculate ΔH for $\text{H}_2 + \text{Cl}_2 \rightarrow 2\text{HCl}$.
A+	7(a) For $\text{CoCl}_2 \cdot 6\text{H}_2\text{O} \rightleftharpoons \text{CoCl}_2 + 6\text{H}_2\text{O}$, predict and explain the effect of (a) adding water, (b) removing water, and (c) heating.	OR	7(b) Define chemical equilibrium and state the conditions needed for a reversible reaction to establish equilibrium in a closed system.
A+	8(a) Define Bronsted-Lowry acid and base. Identify the acid and base in $\text{NH}_3 + \text{H}_2\text{O} \rightleftharpoons \text{NH}_4^+ + \text{OH}^-$.	OR	8(b) Differentiate strong and weak acids using their degree of dissociation; write aqueous dissociation equations for HCl and CH_3COOH .
A+	9(a) From the electronic configuration 2,8,7, identify the group, period, valence electrons and ion usually formed.	OR	9(b) State and explain the trends down Group I for atomic radius, ionization energy and reactivity.
A+	10(a) Complete and explain the possible displacement reactions: $\text{Cl}_2 + 2\text{KBr}$; $\text{Cl}_2 + 2\text{KI}$; $\text{Br}_2 + 2\text{KI}$.	OR	10(b) Describe the trends down Group VII in density and reactivity, and arrange HF , HCl , HBr and HI in increasing thermal stability.

Short-Question Reserve Bank

Prepare these immediately after the ten predicted pairs; each is a credible 3-mark replacement.

Rank	Reserve question
A	Explain allotropic forms of carbon; compare diamond and graphite briefly.
A	Write relative masses and relative charges of electron, proton and neutron.
A	Derive formulas of compounds formed by $\text{Mg}^{2+}/\text{Cl}^-$, $\text{Al}^{3+}/\text{O}^{2-}$ and $\text{Ca}^{2+}/\text{NO}_3^-$.
A	Explain the effect of temperature on solubility of an endothermic and an exothermic solution.
A	Write equations for reactions of HCl with Zn, NaOH and Na_2CO_3 ; state observations.
A	State three characteristic properties/industrial catalytic uses of transition elements.
A	State sources and harmful effects of CO, particulates, NO_x , SO_2 and methane.
A	Explain why nitrates/phosphates in water are harmful and state their sources.
A	Explain sedimentation, filtration, activated carbon treatment and chlorination of domestic water.
A	Explain water scarcity in Pakistan and suggest three practical control measures.
A	State the role of N, P and K in NPK fertilizer; name three nitrogen-containing fertilizers.
A	Identify functional groups in alcohol, aldehyde, ketone, carboxylic acid, amine, ester and amide.
A	Define homologous series and list three characteristics.
A	Define substitution reaction and write the photochemical chlorination of methane.
A	Write balanced equations for complete and incomplete combustion of methane.
A	Name carbohydrates, proteins and lipids; give two sources and one function of each.

SECTION C — Predicted Long Questions

Marks: 18 • Attempt one side of each numbered pair. Each part carries 6 marks.

A+	Q3(a) Using electronic configurations, explain the trends across a period and down a group for atomic radius, ionization energy and electronegativity. Apply the trends to compare Na/K and F/Cl.	OR	Q3(b) Draw dot-and-cross diagrams for NaF, MgO and H ₂ O. Identify each bond and explain electrical conductivity and melting-point behaviour from bonding and structure.
A+	Q4(a) Explain the Bronsted-Lowry concept of acids and bases. Write balanced reactions of HCl with Zn, NaOH and Na ₂ CO ₃ , naming the products and observations.	OR	Q4(b) Define alkanes and explain why they are saturated and generally unreactive. Describe preparation by hydrogenation of an alkene/alkyne and reduction of an alkyl halide, with balanced equations.
A+	Q5(a) Define exothermic and endothermic reactions. Draw and label both energy-profile diagrams (reactants, products, activation energy and ΔH), and explain the effect of a catalyst.	OR	Q5(b) Explain how CO ₂ and CH ₄ cause the greenhouse effect. Describe formation/effects of acid rain from SO ₂ and NO _x and give practical control strategies, including catalytic converters and flue-gas desulfurization.

Long-Question Reserve Bank

1. Calculate mole, mass and number of particles in a multi-part numerical using Avogadro's constant; include molar-mass calculation.
2. Determine relative atomic mass from isotope abundances and discuss carbon dating/medical imaging uses of isotopes.
3. Identify oxidizing/reducing agents by oxidation-number changes in two reactions and explain the redox process.
4. Compare diamond and graphite in structure, bonding, hardness, melting point and electrical conductivity; relate each to industrial use.
5. Differentiate compounds and mixtures, and compare solution, colloid and suspension using composition, particle size/settling and separation.
6. Explain domestic water treatment, pollutants, water-borne diseases and solutions to water scarcity in Pakistan.
7. Explain structural isomerism using C₄H₁₀ and C₄H₈; identify functional groups and characteristics of a homologous series.
8. Discuss carbohydrates, proteins and lipids: sources, recommended/healthy intake principles, and major functions in the human body.

Final 24-Hour Revision Checklist

- ☐ Memorize definitions exactly: isotope, mole, activation energy, equilibrium, Bronsted-Lowry acid/base, homologous series, substitution, corrosion and acid rain.
- ☐ Practise at least three weighted-isotope and three mole/particle numericals; show formula, substitution, units and final answer.
- ☐ Practise dot-and-cross diagrams for NaF, MgO, CaCl₂, H₂O, CO₂, HCN and C₂H₄.
- ☐ Learn oxidation-number rules and practise identifying both oxidizing and reducing agents.
- ☐ Draw exothermic/endothermic profiles neatly and label reactants, products, activation energy and ΔH .
- ☐ Learn Group I and VII trends with reasons; practise group/period identification from electronic configuration.
- ☐ Write and balance the high-frequency equations: acid + metal/base/carbonate; halogen displacement; methane combustion/substitution; alkane preparation; acid/base dissociation.
- ☐ Revise air pollutants, greenhouse effect, acid rain, catalytic converters, water treatment/scarcity and NPK fertilizers.
- ☐ Revise functional groups, C₄ structural isomers, homologous series, and sources/functions of carbohydrates, proteins and lipids.

Exam technique: Read the command word. For a 3-mark response, give three distinct marking points unless the question divides marks differently. For a 6-mark response, use headings, balanced equations/diagrams where relevant, and connect every explanation to the property or observation asked.

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For focused revision of FBISE Chemistry SSC-I, 2nd Annual Examination 2026